Landscape generator template

Introduction

The template resembles a minimal Unity project (Unity version 6.preview) with multiple modules. Instead of a single repository containing the entire system to generate a landscape, it was designed modular with multiple repositories. These repositories are cloned in Unity project as illustrated below.

The main libraries are:

- *terrain (main module)* to generate the landscape
- *3dbuildings* to generate the buildings
- *trees* to place trees on top of terrain
- trambusses to enable public transport
- *underground* (in development)

These libraries require some helper libraries which are:
- *geotiff* for elevation of the terrain
- *vectortile* for the different soil types in the terrain
- *utils* for miscellaneous helpers like math and more

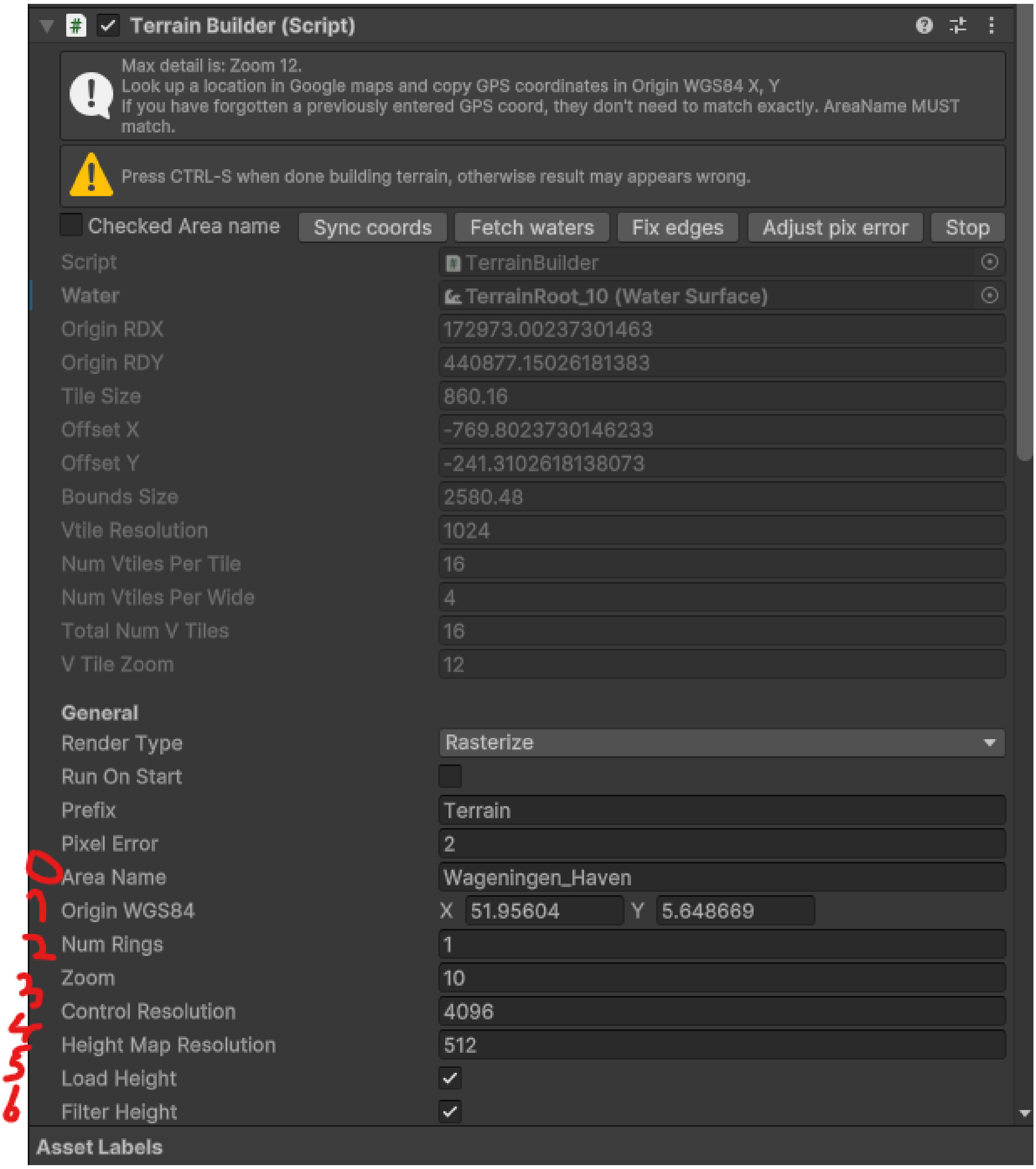
This document will help you set up a landscape with buildings and trees, public transport will be added later as it requires some addition knowledge to get it working.

Example

An area can be generated based on a single GPS-coordinate. Important to remember is that all modules must have the same GPS-coordinate, terrain, buildings, trees, etc. To ensure this the case there is an easy 'Sync' button. The main idea is to work from the Terrain prefab/module. The following image below illustrates this.

The workflow is at follow usually to not interrupt the current scene:

1. Duplicate the original (this scene) (Ctrl-D in Unity on the scene).
2. Set the appropriate WGS84 (GPS) coordinate.
3. Enter a unique name for the area: E.g. Wageningen-campus
4. Check the box on the top left and Press the appearing 'Build' button.
5. Missing tiles (rings can be generated later, simply increase the Num Rings) and Build again.



- *0 Area name*: this where the tiles of the landscape will be generated, as well as the buildings and trees. Both this name as well as the Origin WGS84 (GPS-coordinate) must match in all modules. This is automatically done when the build is started.
  An additional check must first be set to start the building, this is a safety cause to ensure that you do not mix tiles from different areas in the same folder accidentally.

- *1 Origin WGS84*: the GPS-coordinate. Use Google maps to find a location and use r-mouse to see the GPS-coordinate of that location.

- *2 Num Rings:* This is the number of rings of tiles that will be generated. 0 Rings will generate a single tile, 1 ring will generate 9 tiles, 2 rings will generate 25. Tiles = (Rings * 2 + 1)^2.

- *3 Zoom:* This determines the size of the tile and therefore also the quality. The highest zoom (most quality: is 12). The least is 0 (1 tile which covers the entire landscape of Netherlands). **Disclaimer: because the vector tiles are only available at a zoom level 12, the number of vector tiles that need to be processed can increase rapidly. Realize that at a zoom level of 10, 1 single tile is already 16 vector tiles (12 – 10)^2. So for multiple tiles this adds quickly.**

- *4 Control Resolution*: This is the map that tells per pixel what type of soil resides on the terrain. Recommended to leave at 4096. Must be multiple of Heightmap Resolution and not be lower than 128.

- *5 Heightmap Resolution*: This controls the number of points (vertices) in the terrain. Must not be too high, as no such resolution is available anyway and it is not that precise. Recommended to leave at 512 or lower. Small areas may benefit from 1024.

- *6 Load Height / Filter Height*: Whether to load height at all and to apply post filtering. Recommended to leave on. But might be turned of to demonstrate differences or for other purposes.

Example video

See the enclosed sample video that shows step by step how to generate the City of Amersfoort.